SYSTEMS THINKING AND DESIGN - PD2.405

THE EMPTY DESK: Making Shared Lab Resources Readable

Research Lab Commons · SERC × Precog, IIIT Hyderabad · Team **The System Error**

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01 | SYSTEM FRAMING

Describing what the problem is really about · Problem + Boundary

PROBLEM

Two research labs share one floor: **~100 students, 84 desks**, GPUs, 12 lockers, 2 meeting rooms, one fridge, coffee machine and microwave. Students reported occupied-looking empty desks, GPU contention and food waste, yet also cooperative borrowing.

Reframed problem: Shared resources in SERC are not only constrained by limited capacity; they are made harder to access by **fragmented ownership, fragmented visibility, unwritten rules** and dependence on informal negotiation. A resource can be **physically available but still remain effectively unavailable.**

BOUNDARY

OUTSIDE: campus construction · admissions · research evaluation

BOUNDARY: institute rules · cleaning · procurement · admin

INSIDE THE SYSTEM
desks & chairs · local GPU time · fridge, coffee, microwave
lockers & storage · meeting rooms · and the people
who authorise or negotiate their use

Shared location ≠ shared resource (cafeteria vs. its appliances)

CORE TENSION

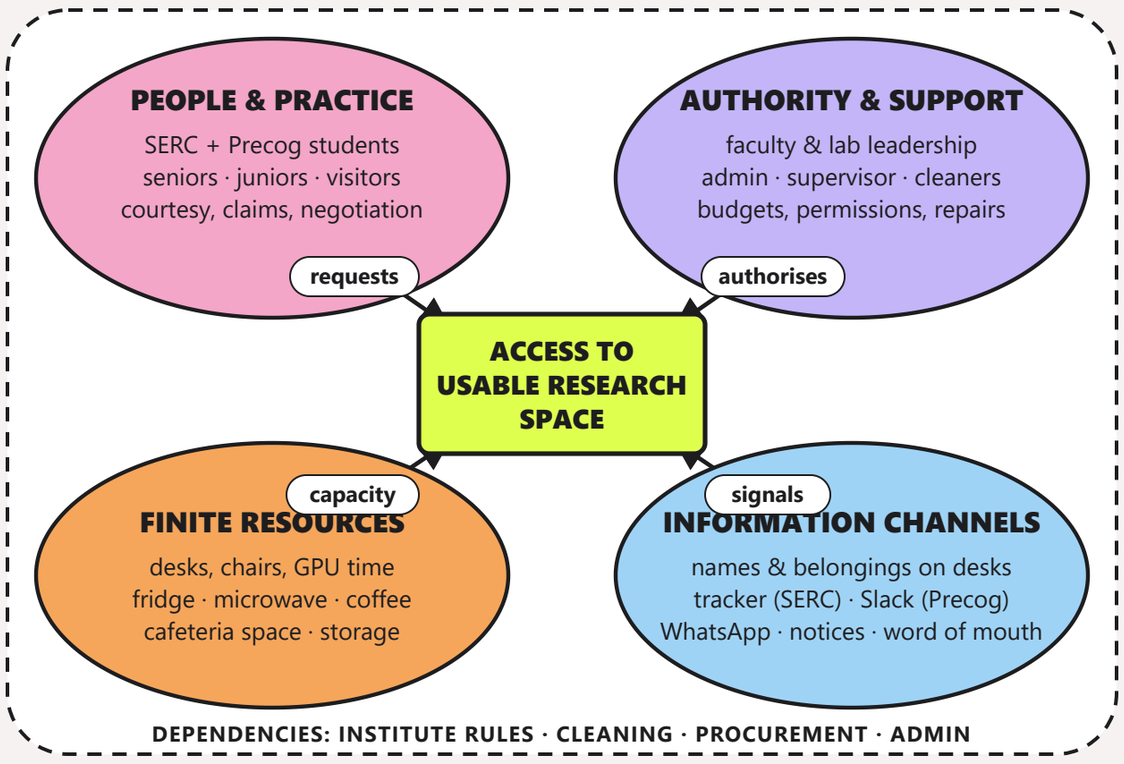
~100	84	2
students	desks	labs, 1 floor

Empty ≠ available.

We found a **readability problem sitting on top of a capacity problem**. Peak-time shortage is real, but newcomers also cannot tell what is free, who holds it, or whom to ask.

02 SYSTEM MAPPING

What makes the system work? · Actors + Relationships · System Map



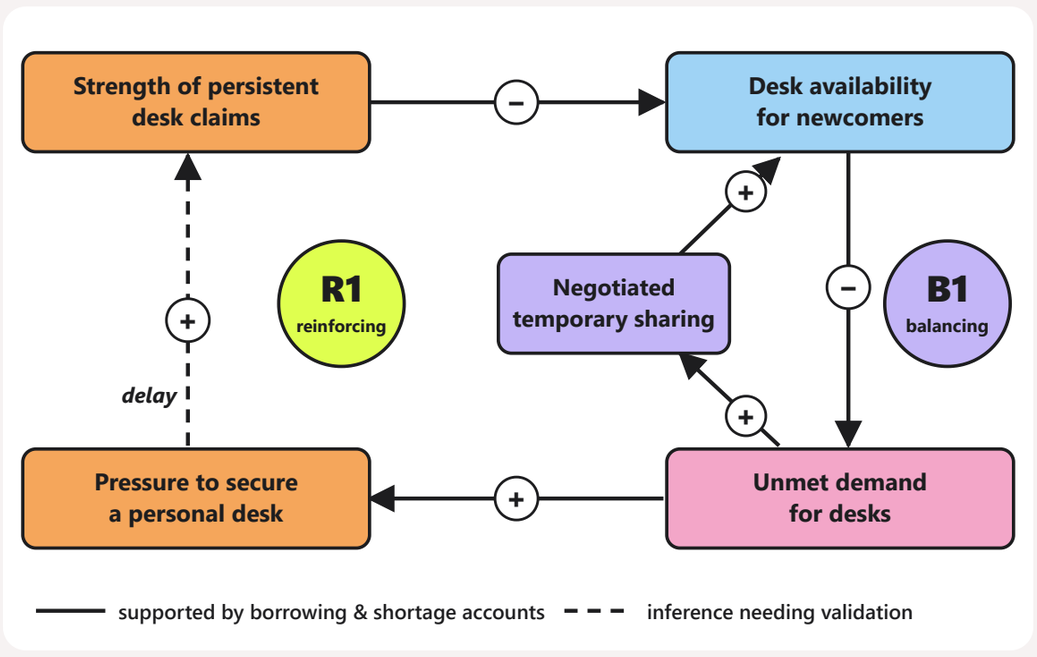
KEY ACTORS & RELATIONSHIPS

Seniors / incumbents	Hold desks and GPU slots; names and belongings act as claims; grant permission informally.
Juniors / newcomers	Must ask to use what looks free; hesitate at desks they were never refused.
Faculty	Control funding and permissions; allocation follows who paid. Act quickly when reached, but small issues rarely reach them.
Student POCs / GPU manager	Keep the SERC tracker; act as informal resource managers and brokers.
Admin, supervisor & cleaners	Room booking, procurement, upkeep, Saturday fridge clear. Report shortages up the chain but never decide eligibility.

No single floor-wide allocation process: only local arrangements held together by courtesy.

03 | SYSTEM DYNAMICS

Why does it persist? · Feedback + Dynamics · CLD + Narrative



R1 CLAIMS REINFORCE SCARCITY

Stronger claims shrink newcomer availability → unmet demand rises → pressure to secure a personal desk grows → over time, claims strengthen further.

B1 NEGOTIATION GIVES RELIEF

Sharing frees desks only when the holder is absent, reachable and willing, and the requester feels able to ask. A junior's hesitation weakens it.

NARRATIVE: WHY IT PERSISTS

Delays hide the loop: **absence → desk release**, **GPU reservation → actual use**, **minor problem → formal awareness**. Because status is unreadable (SERC tracker, Precog Slack, names on desks: all separate), every check becomes a person-to-person query. Problems surface late, so correction is blunt, e.g. **coffee access withdrawn from all students**, and the shared benefit is lost. Underneath sit mental models: *"a desk someone regularly uses is effectively theirs."*

M1 claims persist · M2 status unreadable · M3 access runs on asking · M4 late, blunt correction · M5 allocation follows history

04 | LEVERAGE POINTS

Where can we intervene to create meaningful change? · Intervention Opportunities

▲ **SHALLOW · fast, easy, lower leverage**

LP1	**Correction targeting** SHALLOW	Correct the responsible individual, not the whole group.
LP2	**Loop gain on hoarding** SHALLOW → MEDIUM	Make idle holding visible so it carries a social cost.
LP3	**Information flow ★** CORE · HIGHEST BUILDABLE	One shared visibility layer: holder, active/idle, time held, expected release, contact.
LP4	**Rules of engagement** MEDIUM → DEEP	Light explicit rules: reservation expiry, idle-release, escalation path.
LP5	**Self-organisation** DEEP STRUCTURAL	Standing semesterly reallocation review on current headcount + usage.
LP6	**System goal** DEEP	Utilisation with attribution, not exclusivity.
LP7	**Mental model** DEEPEST	Keep attribution without turning it into exclusivity.

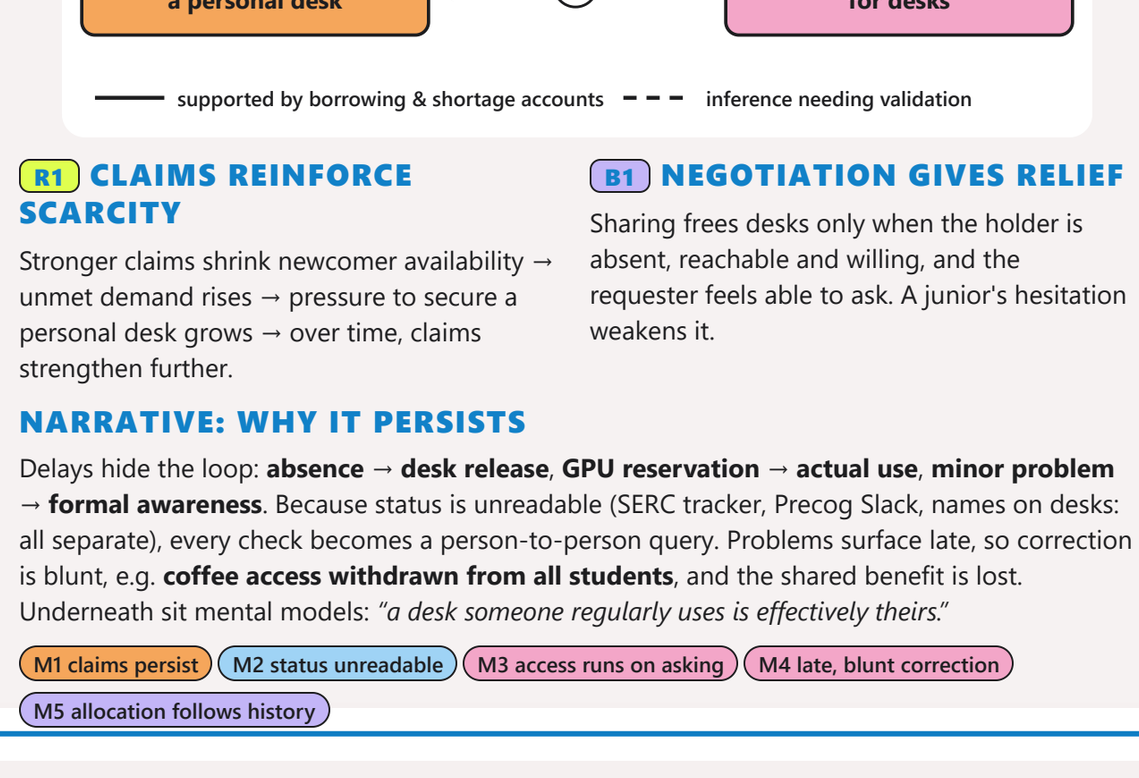
▼ **DEEP · slower, harder, higher leverage**

LP3 → LP4 → LP5 **Highest practical leverage now.** You cannot enforce a rule no one can see: visibility (LP3) enables fair rules (LP4), which enable a self-correcting review (LP5).

Goal shift (LP6): from *"protect what my group funded"* to **"use idle capacity, with attribution"**. Not "remove ownership".

05 | SYSTEMIC DESIGN INTERVENTION

Systemic Design Concept: one system, five parts (not an app)



MECHANISM → INTERVENTION → EXPECTED EFFECT

M2 status unreadable	1 Claim Signals	Free capacity reads as free; fewer manual checks.
M1 claims persist	2 Default-Decay	Claims track actual use; less defensive reserving.
M3 access runs on asking	4 Stewards + ladder	Asking becomes cheaper and less personal.
M4 blunt correction	3 Proportionate Care	Misuse controlled; amenities stay open to all.
M5 history-based allocation	5 Termly Review	Allocation adapts to use before crises.

Kept as is: ask-first norm · faculty authority · equipment attribution · GPU quotas. *Proposed design, low-tech (paper cards, existing GPU page, a shared sheet). Effects predicted, not yet measured.*

PROJECT DOCUMENT

PROJECT CASE STUDY

Faculty : Prof. Raman Saxena